Bisection Method

$f(x) \longrightarrow [low, high] \longrightarrow f(Low) * f(high) < 0$

$$mid = \frac{low + high}{2}$$

if $f(low) * f(mid) < 0$

$[low, mid]$

elif $f(low) * f(mid) > 0$

$[mid, high]$

else return mid;
~~elif~~
~~else~~

$$error = \left| \frac{\text{New mid} - \text{Prv mid}}{\text{New mid}} \right| \times 100\%$$

Advantages

1) Always convergent.

Drawbacks:

1) Slow convergence.
11) issues occur in Positive function and the functions having singularity.

Newton-Raphson Method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\text{error} = \left| \frac{x_{n+1} - x_n}{x_{n+1}} \right| \times 100\%$$

Drawbacks

1) Divergence at inflection points.

2) Division by 0.
   ↳ may not converge.

3) Oscillations near local maxima & minima.

4) Root jumping.
   → $f(x)$ is ~~a~~ oscillating and has a number of roots, ~~one may~~ the initial guess ~~to~~ close to a particular root may converge to some other root.

Naive Gaussian Elimination

pitfalls

1) Division by 0
2) Round-off error
   ↳ solve → Using more significant digits

To solve both use partial pivoting.

Determinant:

k = num of pivoting

① $(-1)^{k} \times \prod_{i=1}^{n} a_{ii}$

② $\det(A) = \dfrac{1}{\det(A^{-1})}$

if $\det(A) \neq 0$ then,

→ A is invertible

→ $AX = C$ has unique solution.

→ $AX = 0$ $X = 0$

Newton's Divided Difference Interpolation

Given $(n+1)$ data points $(x_0, y_0) \cdots (x_n, y_n)$

n^{th} order polynomial is.

$$f_n(x) = \sum_{i=0}^{n} \left(b_i \times \prod_{j=0}^{i-1} (x - x_j) \right)$$

$$= b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2) \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

<u>Linear:</u>

$$f_1(x) = b_0 + b_1(x - x_0)$$

$$\Rightarrow f_1(x_0) = \boxed{y_0 = b_0}$$

$$f_1(x_1) = y_1 = b_0 + b_1(x_1 - x_0)$$

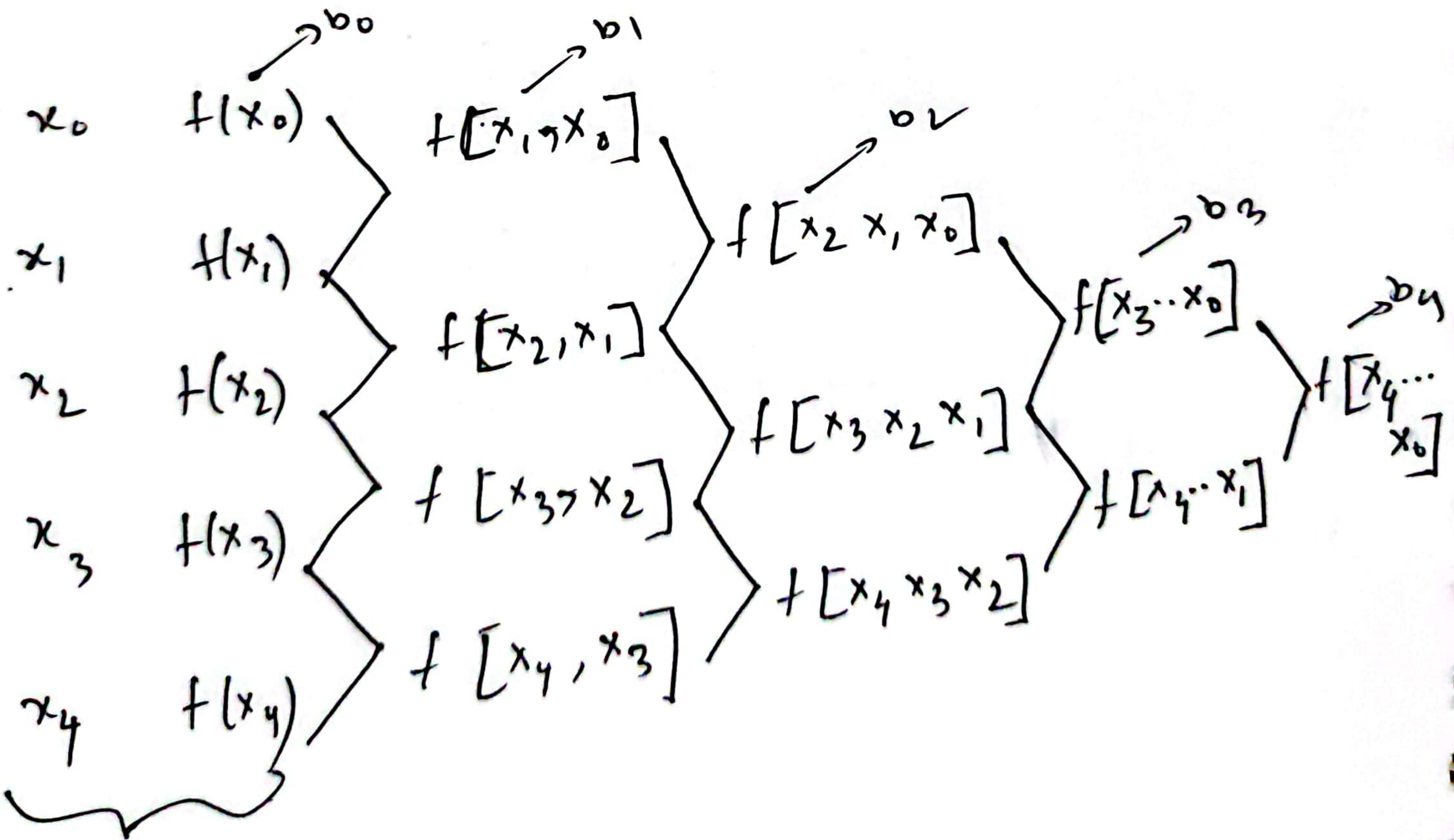
$$\boxed{b_1 = \frac{y_1 - y_0}{x_1 - x_0}}$$

m^{th} divided difference

$$f[x_m, x_{m-1} \cdots x_0]$$

$$= \frac{f[x_m, x_{m-1} \cdots x_1] - f[x_{m-1}, x_{m-2} \cdots x_0]}{x_m - x_0}$$

Graphical solution | $f_4(x)$



order doesn't matter

$$f_3(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1) + b_3(x-x_0)(x-x_1)(x-x_2)$$

x	$f(x)$
x_0 10	227.04 $\rightarrow b_0$
x_1 15	362.78
x_2 20	517.35
x_3 22.5	602.97

$f[x_1, x_0] \rightarrow b_1$

$$\frac{362.78 - 227.04}{15-10} = 27.148$$

$f[x_2, x_1]$

$$\frac{517.35 - 362.78}{20-15} = 30.914$$

$f[x_3, x_2]$

$$\frac{602.97 - 517.35}{22.5-20} = 34.248$$

$f[x_2, x_1, x_0] \rightarrow b_2$

$$\frac{30.914 - 27.148}{20-10} = 0.3766$$

$f[x_3, x_2, x_1]$

$$\frac{34.248 - 30.914}{22.5-15} = 0.4445$$

$f[x_3, x_2, x_1, x_0] \rightarrow b_3$

$$\frac{0.4445 - 0.3766}{22.5-10}$$

$$= 5.432 \times 10^{-3}$$

$$f_3(x) = 227.04 + 27.148(x-10) + 0.3766(x-10)(x-15) + 5.432 \times 10^{-3}(x-10)(x-15)\cancel{(x-517.35)}(x-20)$$

Lagrangian Interpolation

Given $(x_0, y_0), (x_1, y_1), \dots (x_n, y_n)$,

Then, Lagrangian interpolation polynomial is,

$$f_n(x) = \sum_{i=0}^{n} L_i(x)\, f(x_i)$$

↓

n^{th} order polynomial

$$L_i(x) = \prod_{j=0 \,\&\, j \neq i}^{n} \frac{x - x_j}{x_i - x_j}$$

↓

weighting function

→ Ques: → will be given a point x and the order n → find y.

→ choose $(n+1)$ points closest to x then apply the formula.

Trapezoidal Rule of integration

[approximates the integral by 1st order polynomial]

For n segment,

$$h = \frac{b-a}{n}$$

$$\therefore \int_a^b f(x)\,dx = \frac{h}{2n}\left[f(a) + 2\sum_{1}^{n-1} f(a+ih) + f(b)\right]$$

True error: $$E_t = -\frac{(b-a)^3}{12} f''(\xi) \qquad a < \xi < b$$

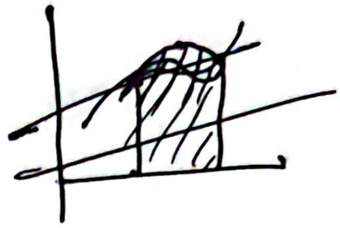
(for single segment)

For n segment

$$E_t = -\frac{(b-a)^3}{12n^2} \times \frac{\sum_{1}^{n} f''(\xi_i)}{n}$$

Simpson's $\frac{1}{3}$rd rule

approximates the integral by 2nd order polynomial



For $n=2$:

$$\int_a^b f(x)\,dx$$

$$= \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

For n segments:

$$\int_a^b f(x)\,dx = \frac{b-a}{3n}\left[f(a_0) + 4\sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(\overset{a+h}{x_i})\right.$$

$$\left. + 2\times\sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(\overset{a+2h}{x_i}) + f(b)\right]$$

$$\# \int_1^2 e^{x^3}\,dx \quad ; \quad n=4$$

$$h = \frac{b-a}{n} = \frac{2-1}{4} = \frac{1}{4} = 0.25$$

$$\int_1^2 e^{x^3}\,dx = \frac{2-1}{3\times 4}\Big[f(1) + 4f(1.25) + 2f(1.5) + 4f(1.75) + f(2)\Big]$$

$$= \frac{1}{12}\Big[e + 4\times 7.0507 + 2\times 29.2243 + 4\times 212.5920 + 2980.9579\Big]$$

$$= 326.7246$$

Error

Error is single segment:

$$E_t = -\frac{(b-a)^5}{2880} f^4(\xi)$$

In multiple segment:

$$E_t = -\frac{(b-a)^5}{180n^4} \times \frac{\sum_{1}^{n/2} f^4(\xi_1)}{n/2}$$

Linear Regression

Least Square Criterion is used.

$$S_r = \sum_{1}^{n} E_i^2 = \sum_{1}^{n} (y_i - a_0 - a_1 x_i)^2$$

$$\frac{\partial S_r}{\partial a_0} = -2 \sum_{1}^{n} (y_i - a_0 - a_1 x_i) = 0$$

$$\frac{\partial S_r}{\partial a_1} = \cancel{-2a x_i \sum} \; -2 \sum_{1}^{n} x_i (y_i - a_0 - a_1 x_i) = 0$$

$$\boxed{a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}}$$

$$\boxed{a_0 = \frac{\sum y_i - a_1 \sum x_i}{n}}$$

minima test:

If function $f(x,y)$ has a critical point (a,b) from 1st derivative test, then (a,b) is a minima if,

$$\frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 > 0 \quad \text{and}$$

$$\boxed{\frac{\partial^2 f}{\partial x^2} > 0 \quad \text{or} \quad \frac{\partial^2 f}{\partial y^2} > 0}$$

Polynomial Regression

$$f_n(x)\ \cancel{y} = a_0 + a_1 x + a_2 x^2 \cdots + a_n x^n$$

$$S_r = \sum_1^n \left(y_i - a_0 - a_1 x_i - a_2 x_i^2 \cdots - a_n x_i^n\right)^2$$

$$\Downarrow \quad \frac{\partial S_r}{\partial a_0} = \frac{\partial S_r}{\partial a_1} = \frac{\partial S_r}{\partial a_2} \cdots \frac{\partial S_r}{\partial a_n} = 0$$

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \cdots & \Sigma x_i^n \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \cdots & \Sigma x_i^{n+1} \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \cdots & \Sigma x_i^{n+2} \\ \vdots & \vdots & \vdots \ddots & \vdots \\ \Sigma x_i^n & \Sigma x_i^{n+1} & \Sigma x_i^{n+2} \cdots & \Sigma x_i^{2n} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma y_i x_i \\ \Sigma y_i x_i^2 \\ \vdots \\ \Sigma y_i x_i^n \end{bmatrix}$$

For 2nd order:

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma y_i x_i \\ \Sigma y_i x_i^2 \end{bmatrix}$$